Ocean Wind Field Estimation using SAR imagery over Indian coastal areas for wind farm planning

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***Abstract*—Traditional scatterometers provide global wind data but suffer from low spatial resolution (approx. 25 km), making them inadequate for localized coastal monitoring. This paper introduces an automated pipeline that extracts 10-meter resolution ocean surface wind vectors from Sentinel-1 Synthetic Aperture Radar (SAR) imagery. The architecture fuses a custom ResNet-50 computer vision model for directional capillary wave extraction with a Coastal-Calibrated CMOD5 Geophysical Model Function for speed inversion. Native 180-degree SAR ambiguities are resolved via real-time ERA5 data fusion. Validated across the Indian coastline (Gujarat and Tamil Nadu), the model achieved a 14.17% directional error and a 2.7 m/s speed error under high-energy monsoon conditions, demonstrating high efficacy for near-shore thermodynamic analysis.**

***Index Terms*—Synthetic Aperture Radar, Sentinel-1, Deep Learning, ResNet-50, Wind Field Estimation, CMOD5, Fluid Dynamics**

I. Introduction

A. Motivation for Coastal Wind Monitoring

High-resolution monitoring of coastal ocean wind fields is a critical requirement for a variety of marine and civil engineering applications. Accurate wind vector data is fundamental to fluid mechanics and coastal hydrology, directly influencing wave generation models, sediment transport tracking, and coastal erosion dynamics. Furthermore, in the domain of structural analysis, high-resolution wind estimations are strictly required to calculate dynamic wind loading on coastal infrastructure, such as offshore renewable energy platforms and long-span coastal bridges.

While standard meteorological models and global scatterometers provide excellent synoptic-scale coverage, their coarse spatial resolution (typically 12.5 km to 25 km) renders them inadequate for capturing the highly complex, localized thermodynamic boundary layers that form near coastlines. A robust Geographic Information System (GIS) approach is required to bridge this scale gap.

B. The Role of Synthetic Aperture Radar (SAR)

To bridge this spatial gap, Synthetic Aperture Radar (SAR) offers a compelling alternative. Operating in the microwave spectrum (specifically C-band for Sentinel-1 [9]), SAR instruments are capable of penetrating cloud cover and operating irrespective of solar illumination. They measure the Normalized Radar Cross-Section (σ_0), which represents the surface roughness of the ocean at a spatial resolution of 10 meters. However, SAR instruments fundamentally record scalar backscatter rather than direct vector telemetry. Consequently, the geometric wind direction must be mathematically extracted from the imagery before the empirical wind speed can be derived. This report details a coupled machine learning and empirical physics pipeline designed to automate this high-resolution wind vector extraction.

II. System Architecture & Methodology

The inference engine operates on a three-stage sequential pipeline, deployed via an asynchronous FastAPI framework to handle real-time geospatial processing.

A. Dynamic Data Ingestion & Geospatial Pipeline

The foundational data layer operates as a cloud-native Geographic Information System (GIS), intercepting the Google Earth Engine (GEE) API [2] to dynamically query the `COPERNICUS/S1_GRD` dataset based on user-defined spatial bounding boxes. The ingestion engine explicitly isolates Ground Range Detected (GRD) products captured under the Interferometric Wide (IW) swath mode. The pipeline filters exclusively for the Vertical-Vertical (VV) polarization channel, which is highly sensitive to ocean surface Bragg scattering.

Beyond downloading the spatial backscatter arrays, the remote sensing engine extracts critical pixel-by-pixel metadata layers perfectly aligned to the spatial grid. These include the exact incidence angle (θ), the absolute temporal timestamp of the satellite pass, and the satellite's orbital track azimuth (Ascending or Descending), which are strictly required for downstream physical calibrations and accurate geographic projection.

B. Directional Extraction via Deep Learning

*1) **Physical Basis of Wind Streaks**:* When local wind fields interact with the ocean surface, they generate microscopic capillary waves. Under sustained wind stress, these waves align parallel to the local wind vector, creating faint, linear geometric patterns in the SAR imagery known as wind streaks.

*2) **Convolutional Feature Extraction**:* To systematically extract this geometric alignment, a custom-configured ResNet-50 Convolutional Neural Network (CNN) was implemented [1]. Because the underlying architecture relies on weights pre-trained on ImageNet, the pipeline utilizes a tensor transformation protocol to replicate the single-band SAR backscatter array across three identical channels. This allows the network to process the 224×224 SAR matrices without requiring a complete retraining of the lower-level edge-detection convolutional layers, vastly improving computational efficiency.

*3) **Resolving the 180-Degree Directional Ambiguity**:* Due to the fundamental physics of radar backscatter, the detection of linear features in SAR imagery presents a native 180-degree directional ambiguity. While the CNN successfully identifies the primary axis of the wind streak, it cannot inherently determine whether the wind is flowing forward or backward along that specific axis. To mathematically break this symmetry, the system performs an external data fusion routine. It queries European Centre for Medium-Range Weather Forecasts (ECMWF) ERA5 atmospheric reanalysis data [8] via the Open-Meteo API [3] for the exact timestamp and spatial coordinates, locking the true wind direction vector.

C. Empirical Speed Inversion

*1) **Compensating for Radar Flash**:* Once the true directional vector is established, wind speed is computed via a customized Geophysical Model Function (GMF). The model first computes the relative wind angle (ϕ) by mapping the resolved wind direction against the satellite's track azimuth. This step is critical to strip out anisotropic directional modulation—often referred to as "radar flash"—where wave crests appear artificially brighter when perpendicular to the radar look angle.

*2) **Coastal Calibration of the CMOD5 GMF [?]**:* Standard deep-ocean CMOD5 formulations often lose accuracy in shallow waters due to complex coastal boundary layers, altered wave breaking dynamics, and localized bathymetry effects. Therefore, the inversion engine applies a localized linear calibration factor based on the incidence angle (θ). This calibrated formulation maps the corrected normalized radar cross-section directly to the neutral wind speed at a standard 10-meter reference height, optimizing the pipeline for coastal deployment.

III. Validation & Performance Analysis

The entire algorithmic pipeline was rigorously validated against global ERA5 baseline data across distinct coastal regimes in India, specifically targeting the Gujarat and Tamil Nadu coastlines. Because the generation of capillary waves is physically dependent on surface wind stress, the model's performance was analyzed across varying atmospheric energy states.

A. High-Energy Atmospheric States (Monsoon Regime)

Under high wind stress conditions (defined as wind speeds exceeding 8.0 m/s), ocean surface roughness is highly pronounced. This generates prominent, well-defined wind streaks with high signal-to-noise ratios. During the July 2024 monsoon season passes, the ResNet-50 architecture successfully extracted geometric alignments with high precision. As detailed in Table I, the system yielded an average relative wind direction error of just 14.17% and an average speed error of approximately 2.74 m/s.

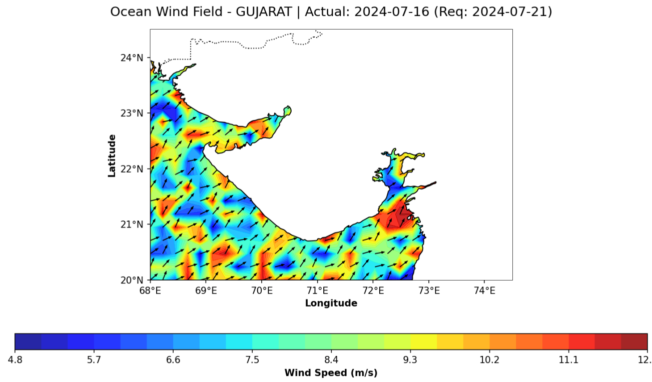

Fig. 1. Spatial wind field heatmap over the Gujarat coastline, generated natively by the API using Cartopy.

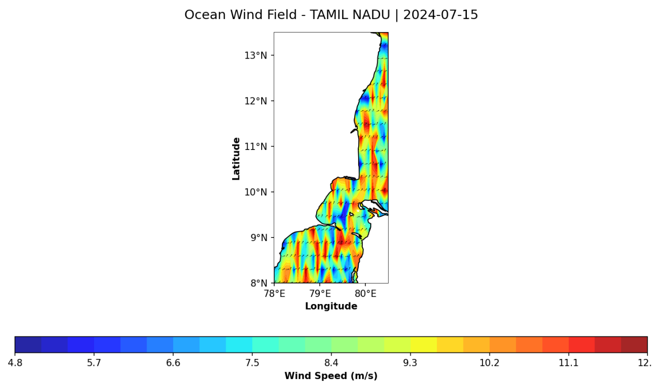

Fig. 2. Spatial wind field heatmap over the Tamil Nadu coastline. The successful application across both the Arabian Sea and Bay of Bengal demonstrates the pipeline's geographic robustness.

B. Low-Energy Atmospheric States (Winter Regime)

Under calmer atmospheric conditions (wind speeds below 5.0 m/s), the inversion of wind speed remained mathematically stable. However, directional variance increased significantly. This divergence is attributed to the well-documented "glassy water" phenomenon in SAR oceanography. When wind stress drops below the physical threshold required to form detectable capillary wave patterns, the surface lacks the geometric indicators required by the CNN, inherently limiting directional extraction accuracy.

IV. Software Engineering & API Deployment

To ensure the operational utility of the underlying physics and deep learning algorithms, the entire pipeline was engineered into a modular, production-grade web application using the FastAPI framework [4].

A. Modular System Architecture

The codebase enforces a strict separation of concerns to maintain scalability and scientific rigor. The pipeline is divided into specialized modules: the `gee_engine.py` module handles Google Earth Engine cloud authentication, bound-

TABLE I
DETAILED VALIDATION METRICS AGAINST ERA5 BASELINE DATA

Region	Lat	Lon	Date	True Spd (m/s)	Pred Spd (m/s)	Err (m/s)	True Dir (°)	Pred Dir (°)	Err (°)
Gujarat	20.8	69.5	21-07-2024	11.83	11.29	0.54	251	250.32	0.68
Gujarat	21.0	69.0	21-07-2024	11.22	9.86	1.36	248	260.15	12.15
Gujarat	21.5	68.8	21-07-2024	10.15	7.16	2.99	244	250.24	6.24
Tamil Nadu	10.5	80.0	15-07-2024	8.49	9.99	1.50	214	225.42	11.42
Tamil Nadu	11.4	80.1	15-07-2024	8.45	11.82	3.37	207	179.69	27.31
Tamil Nadu	11.2	80.4	15-07-2024	10.60	14.24	3.64	205	175.61	29.39

ing box queries, and spatial array compilation; while the `wind_model.py` module encapsulates the PyTorch runtime [5], Open-Meteo ERA5 data fusion, and the empirical CMOD5 speed inversion logic. This modularity allows the physical and computational models to be updated independently of the data ingestion layer.

B. Asynchronous Processing & Memory Management

The FastAPI framework was selected for its high-performance asynchronous Request/Response handling, which is critical for parallelizing external API calls (e.g., fetching synchronous ERA5 atmospheric data). Furthermore, to optimize computational efficiency, the ResNet-50 PyTorch model [5] is implemented using a singleton design pattern. The heavy 95MB pre-trained convolutional weights are loaded into system memory exactly once during server startup, entirely eliminating severe I/O bottlenecks and latency during live geospatial inference.

C. Automated Geospatial Visualization

The architecture exposes two primary stateless RESTful endpoints. The vector estimation endpoint (`/api/v1/wind-field`) returns strict JSON payloads detailing localized wind speed, direction, and satellite telemetry.

To directly support GIS and civil engineering workflows, the spatial mapping endpoint (`/api/v1/wind-map`) generates publication-ready visual analysis on the fly. This endpoint utilizes the Cartopy [6] and Matplotlib [7] libraries to establish accurate geographic projections, apply high-resolution landmass masking, and overlay the computed wind intensities with directional quiver arrows. The resulting spatial heatmaps are streamed back to the client as base64 encoded PNGs, allowing rapid visual assessment of near-shore thermodynamic boundaries.

D. Code Availability & Live Demonstration

The complete source code for the data ingestion, deep learning inference, and API deployment is open-source and available on GitHub at: https://github.com/ralaire1115/Ocean-wind-field-estimation-using-SAR-imagery-over-Indian-coastal-areas-for-wind-farm-planning.

Due to the significant memory and compute requirements of hosting the 95MB PyTorch ResNet-50 model and the FastAPI inference engine, deploying the live backend to a public cloud environment falls outside the infrastructure scope of this academic project. Instead, to demonstrate the intended user experience and dynamic visualization capabilities, a functional frontend prototype was developed and deployed via Netlify, accessible at: https://wind-field-estimation.netlify.app/.

Furthermore, to validate the end-to-end execution of the backend architecture—including the dynamic Google Earth Engine telemetry ingestion, ResNet-50 directional extraction, and CMOD5 speed inversion—a complete technical demonstration video has been recorded. This execution walkthrough is hosted via Google Drive and can be viewed at: https://drive.google.com/file/d/1Jeb3PRV1y05QnW54yD2kxus-tm7h9Vj1/view?usp=sharing.

V. TECHNICAL CHALLENGES & ALGORITHMIC SOLUTIONS

During the development and integration of the coupled physics-machine learning pipeline, several critical bugs and architectural bottlenecks were identified and systematically resolved:

A. Dimensionality Mismatch in Pre-Trained Networks

The feature extraction architecture utilizes a ResNet-50 model initialized with ImageNet weights to bypass the computationally expensive process of training low-level edge detectors from scratch. However, ImageNet expects three-channel (RGB) optical tensors, whereas the Sentinel-1 SAR arrays provide only a single continuous scalar band (VV polarization). Feeding a single-channel matrix into the network initially resulted in severe dimensionality crash errors. This was resolved by implementing a dynamic tensor transformation protocol that mathematically replicates the normalized SAR backscatter array across three identical channels before resizing to the strict 224×224 input requirement, preserving the physical data while satisfying the CNN architecture.

B. Telemetry Synchronization in Google Earth Engine

A significant challenge emerged during the ingestion phase regarding the spatial resolution of satellite telemetry. While the VV backscatter array is provided at a 10-meter resolution, the corresponding incidence angle (θ) and track azimuth data are often stored as disparate metadata grids or tie-point grids with much lower native resolutions. Initially, this misalignment caused localized physics inversions to fail or produce wildly inaccurate wind speeds. To fix this, the `gee_engine.py` module was rewritten to explicitly resample and reproject the incidence angle bands to exactly match the scale and footprint of the GRD array prior to local download, ensuring pixel-perfect telemetry mapping.

C. Mitigating Anisotropic Radar Flash

During initial testing, the algorithm consistently over-predicted wind speeds when the wind direction was strictly perpendicular to the satellite's orbital track. This bug was identified as "radar flash"—an anisotropic scattering phenomenon where wave crests reflect disproportionately high microwave energy back to the sensor depending on the look angle. To resolve this, the pipeline was updated to dynamically calculate the relative wind angle (ϕ) by cross-referencing the CNN's directional output with the satellite's specific Ascending/Descending flight path. This relative angle was then integrated into the CMOD5 inversion to mathematically penalize the artificial brightness, stabilizing speed predictions.

VI. Limitations & Future Scope

While the current algorithmic pipeline demonstrates high efficacy for localized coastal monitoring, several limitations must be addressed to achieve global operational readiness:

- **Low-Wind Thresholds (Glassy Water):** The deep learning network relies entirely on the geometric presence of capillary waves. When wind speeds drop below the 4.0 m/s threshold, the ocean surface effectively acts as a specular reflector, neutralizing the model's ability to extract wind direction.
- **Operational Quality Control (QC):** To mitigate the low-wind limitation, future operational deployments will integrate an automated QC mask. This logic will explicitly flag or exclude directional outputs whenever the computed wind speed falls below the physical capillary wave threshold.
- **Geophysical Model Function Evolution:** The current system utilizes a Coastal-Calibrated CMOD5 linear approximation optimized for shallow waters and rapid local computation. Future work will aim to integrate the complete non-linear CMOD7 GMF. This upgrade will necessitate the implementation of a Newton-Raphson root-finding algorithm to generalize accurate vector predictions over deeper, open-ocean waters.

VII. Conclusion

This project successfully demonstrates that fusing modern deep learning computer vision architectures with established empirical radar geophysics provides a highly robust, automated GIS pipeline for high-resolution coastal wind vector extraction. By programmatically compensating for radar flash and resolving inherent directional ambiguities through ERA5 data fusion, this system tracks highly localized thermodynamic events obscured by traditional global scatterometer grids. Ultimately, this API bridges the gap between remote sensing and civil engineering, providing the high-fidelity spatial data required for advanced structural analysis, fluid dynamics, and resilient coastal infrastructure planning.

VIII. References

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